Cytochrome P450 Inhibitor Classification with Statistical Learning

by
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A Thesis Submitted to
The Faculty of the Graduate School at
North Carolina Central University
In Partial Fulfillment of the Requirements
for the Degree of Master of Science

Durham
2015

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Abstract

SIEGLE, DANIEL E, M.S. "Cytochrome P450 Inhibitor Classification with Statistical Learning (2015)" Directed by Dr. Weifan Zheng, 55pp.

This project compares methods for predicting cytochrome P450 (CYP) inhibition from compound structure. CYPs are a natural first target for *in silico* models because adverse drug-drug interactions caused by CYP inhibition are a leading reason for rejecting drug candidates. I built and compared several non-linear, high-dimensional classification models using a large, publicly available high-throughput luminescence assay (PubChem AID 1851) against five CYP isozymes (1A2, 2C9, 2C19, 2D6 and 3A4). The methods compared are a Bayesian Binary QSAR method from the Molecular Operating Environment (MOE) and three standard machine learning methods implemented in Python: k-nearest neighbors, random forests and support vector machines. All methods performed well, with test set accuracies of 0.68 to 0.81. On every isozyme, all three Python methods were more accurate than MOE Binary QSAR on the test set, and support vector machines were the most accurate. Freely available software can therefore match or exceed a method widely accepted in industry.

Acknowledgements

To my advisor and committee chair Dr. Weifan Zheng for all his guidance and support. Thanks also to committee members Dr. ClarLynda Williams-DeVane and Dr. Tomas Ding for their contributions and consideration.

I would also like to thank Dr. Johnathan Sexton for advising my first foray into research and special thanks to Natacha Janvier-Derilus for unflagging professionalism in advising and dedication to the students of the BRITE program.

Contents

List of Tables **iv**

List of Figures **v**

1 Background **1**
1.1 Drug Discovery Productivity Challenges 1
1.2 Cytochrome P450 Superfamily 3
1.3 Early Compound Profiling *In Silico* 7
1.4 Quantitative Structure-Activity Relationship 9
1.5 Statistical Machine Learning 12
1.6 Sources of Data for Learning 13
1.7 Reproducibility 15
1.8 Aims 17

2 Materials and Methods **18**
2.1 Review of PubChem AID 1851 18
2.2 Data Set Preparation and Molecular Descriptor Generation 20
2.3 Data Preprocessing 22
2.4 Modeling - Mapping Descriptors to Activity Data 23
2.4.1 Binary QSAR in the Molecular Operating Environment 23
2.4.2 k-Nearest Neighbors (k-NN) 25
2.4.3 Random Forest (RF) 25
2.4.4 Support Vector Machines (SVM) 25
2.5 Model Evaluation and Validation 26

3 Results **31**
3.1 1A2 MOE Models 32
3.2 2C9 MOE Models 33
3.3 2C19 MOE Models 34
3.4 2D6 MOE Models 35
3.5 3A4 MOE Models 36
3.6 1A2 Classification Method Comparison 37
3.7 2C9 Classification Method Comparison 37
3.8 2C19 Classification Method Comparison 38

3.9 2D6 Classification Method Comparison 38
3.10 3A4 Classification Method Comparison 39
3.11 Python Classifier Test Set Results 39
3.12 Overall Method Comparison 41

4 Discussion **43**
4.1 Other Attempts at Modeling PubChem AID 1851 43
4.2 Sources of Error 44
4.3 Machine Learning for Pharmaceutical Sciences 45

5 Conclusion **47**

Bibliography **49**

List of Tables

3.1 1A2 MOE Model Results 32
3.2 2C9 MOE Model Results 33
3.3 2C19 MOE Model Results 34
3.4 2D6 MOE Model Results 35
3.5 3A4 MOE Model Results 36
3.6 Comparison of Classification Methods for 1A2 37
3.7 Comparison of Classification Methods for 2C9 37
3.8 Comparison of Classification Methods for 2C19 38
3.9 Comparison of Classification Methods for 2D6 38
3.10 Comparison of Classification Methods for 3A4 39
3.11 Python Classifier Test Set Results 39

List of Figures

1.1 Ribbon Diagram of CYP1A2 4
1.2 Mechanism of Action of Cytochrome P450 Catalytic Heme Group . . 5
1.3 Ribbon Diagram of CYP3A4 with Heme Center 6
1.4 Typical QSAR Workflow 12

2.1 Data Download and Preparation 20
2.2 Isozyme Data Set Preparation 21
2.3 Supervised Classification Model Training and Validation 23
2.4 Supervised Learning Workflow for Classification 30

3.1 1A2 MOE Model Accuracy 32
3.2 2C9 MOE Model Accuracy 33
3.3 2C19 MOE Model Accuracy 34
3.4 2D6 MOE Model Accuracy 35
3.5 3A4 MOE Model Accuracy 36
3.6 ROC Curves on the Test Sets 40
3.7 Inhibitor Classification Method Comparison on the Training Set . . . 41
3.8 Inhibitor Classification Method Comparison on the Test Set 42

CHAPTER 1

Background

1.1 Drug Discovery Productivity Challenges

The goal of pharmaceutical sciences is to identify safe and efficacious drugs for the market. Toxicity is a large contributor to drug candidate attrition in drug development. Inhibition of enzymes in the cytochrome P450 superfamily is a major source of toxicity in human and animal models because of their role in first pass metabolism. If a compound inhibits cytochrome P450, toxic effects are likely to follow if first pass metabolism fails to clear other therapeutic compounds or alters their pharmacokinetics. Because the discovery and development of each new pharmaceutical drug requires more than a billion dollars across an average of 12 years, it would be useful to know whether a compound of interest inhibits cytochrome P450s as early as possible.

The cytochrome P450 superfamily is large and varied. Different isozymes metabolize different substrates, and pharmacokinetic and pharmacodynamic variability within and between individuals can be high, for reasons not fully characterized. Assaying every isozyme and single nucleotide polymorphism variant against every compound of interest is cost-prohibitive. This project tests how well computational methods predict cytochrome P450 inhibition from chemical structure alone.

Drug development productivity has declined for three decades. The difficulty of improving upon current treatments in novel ways, increasingly cautious regulatory environments, and over-reliance on early stage identification of 'silver bullet' treatments that end up as late stage failures have all been proposed as causes [Scannell et al., 2012]. Integrated computational approaches have been proposed as one way to control costs in ongoing pharmaceutical research and development [Visser et al., 2014]. The foundational skills demonstrated in this thesis are needed to pursue a systems approach to drug development that industry has recently turned toward as a way to boost R&D productivity [Berg, 2014]. This thesis is also part of the larger project of *in silico* drug development, which aims to reduce reliance on exploratory *in vivo* and clinical testing, increase the number of effective treatments, and shorten development time.

Quantitative structure-activity relationship (QSAR) modeling, which relates compound structure to bioactivity, dates to at least Hansch [Hansch and Fujita, 1964] and has a long history in *in silico* drug development. QSAR began with directly measured, and later derived, compound features, which were used to build expert systems or statistical models that predict biological activity. Over the same decades, machine learning emerged, using computation in an analogous but more general way: to associate features with results, inputs with outputs. Machine learning algorithms learn to perform tasks by generalizing from examples.

Statistical learning shares machine learning's focus on prediction but emphasizes models and methods that scientists and engineers can interpret and use [James et al., 2013]. Extending statistical learning to the pharmaceutical sciences is an important precursor to systems biology and systems pharmacology modeling.

This project compares different techniques for cytochrome P450 inhibition prediction in the framework of statistical learning. To demonstrate generality and applicability to the pharmaceutical sciences, this study covers the five cytochrome P450 (CYP) isozymes that are involved in metabolism of 90% of all therapeutic drugs

(CYPs 1A2, 2C9, 2C19, 2D6 and 3A4) [Veith et al., 2009].

Because reproducibility is a core principle of the scientific method, the code and data used in this study are version-controlled and publicly available. I aimed to make the analysis fully reproducible by automated means, so that others can validate the results and extend the work.

1.2 Cytochrome P450 Superfamily

One of the largest and most functionally diverse protein superfamilies is the cytochrome P450 family of hemoproteins. At the latest count there were more than 2000 identified cytochrome P450 genomic and cDNA sequences that have been divided into a total of 265 different families [Danielson, 2002]. Cytochromes P450 appear in every kingdom from bacteria to higher eukaryotes. Multiple cytochrome P450 genes can be expressed simultaneously as different isozymes and the number of genes per species is highly variable with a tendency for higher eukaryotes to possess more. Cytochromes P450 constitute the major enzyme family capable of catalyzing the oxidative biotransformation of most drugs and other lipophilic xenobiotics and are therefore of particular relevance for clinical pharmacology. The central role that these ubiquitous proteins play as phase I enzymes in human drug metabolism makes them important to the pharmaceutical industry.

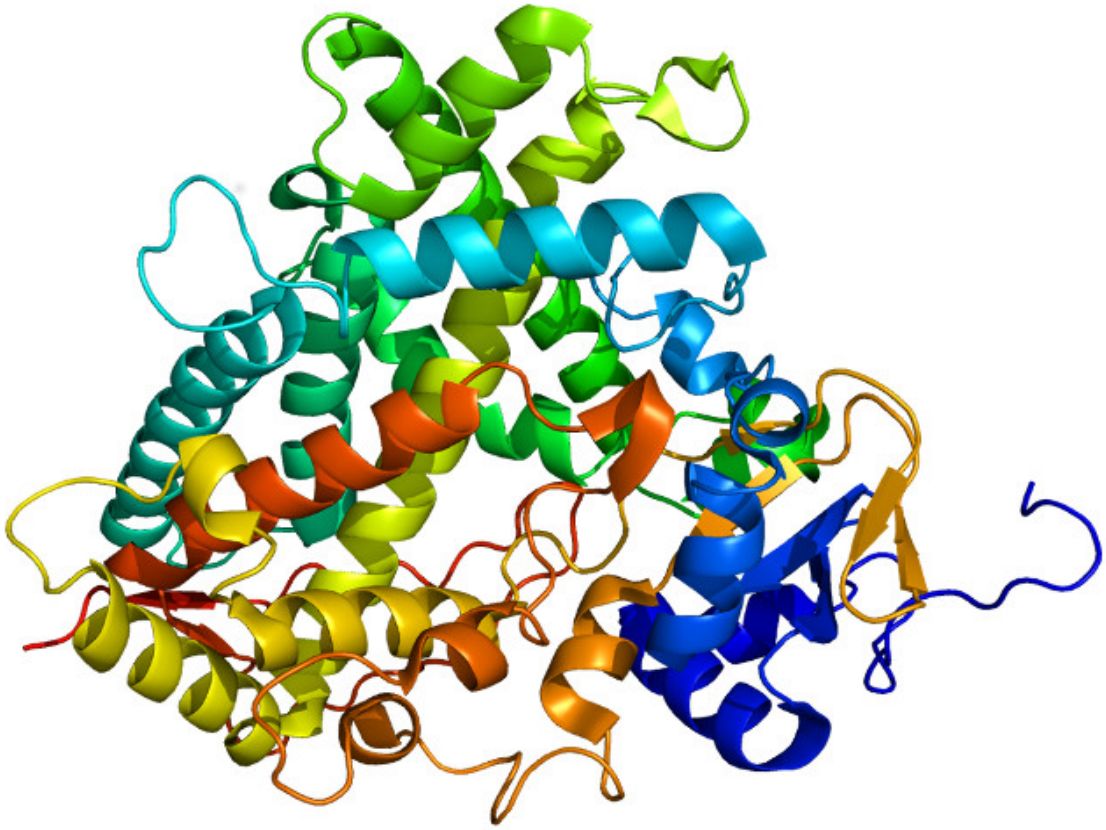


Figure 1.1: Ribbon Diagram of CYP1A2 (PDB 2HI4)

CYP families are classified based on pairwise amino acid sequence identity between individual members. Families CYP 1-3 are involved in phase I metabolism of human drugs and xenobiotic compounds, whereas other CYP families (CYP 4, 11, 17, and 21) are involved in the metabolism of endogenous compounds such as fatty acids, steroids, eicosanoids, bile acids and fat soluble vitamins [Singh et al., 2011].

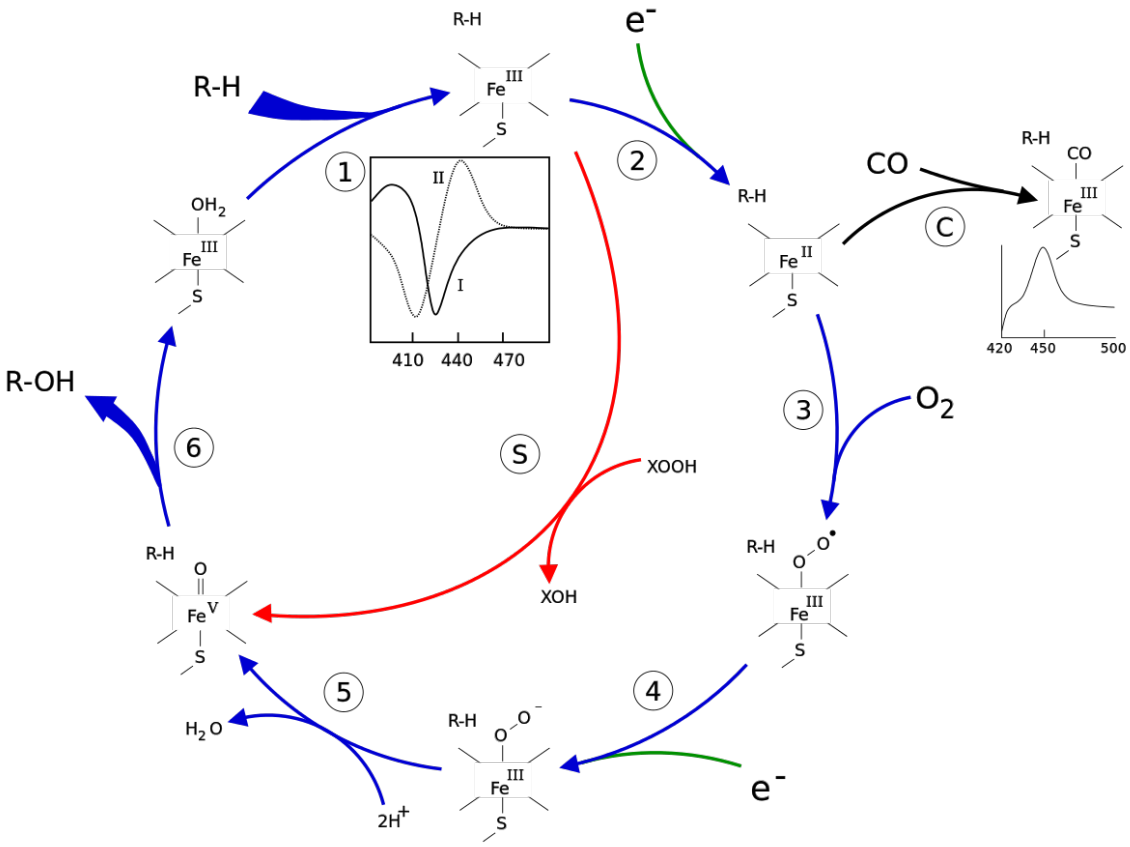


Figure 1.2: Mechanism of Action of Cytochrome P450 Catalytic Heme Group

Figure 1.2 illustrates the catalytic cycle that occurs at the central heme group in all Cytochromes P450 as follows:

1. First the substrate binds and induces a conformational change in the enzyme that usually displaces a water group. This brings the substrate in close proximity to the heme group.

2. Bound substrate induces electron transfer from a nearby reductase that provides NAD(P)H. At this point the -CO complexed with the heme group absorbs light spectra at max 450 nm, noted by the 450 in the original naming of Cytochromes P450.

3. A second oxygen molecule displaces a carbon.

4. A second electron is transferred from a nearby reductase.

5. Rapid double protonation of the peroxo group releases one molecule of water, leaving the ferrous center with one double-bonded oxygen. An alternative route to this chemical state is shown by the S step in Figure 1.2, called the 'peroxide shunt', that entails oxidation of the heme center directly.

6. The reaction catalyzed at this point depends on the substrate and isozyme; the figure shows a hydroxylation, a common way to make lipophilic substances more water soluble.

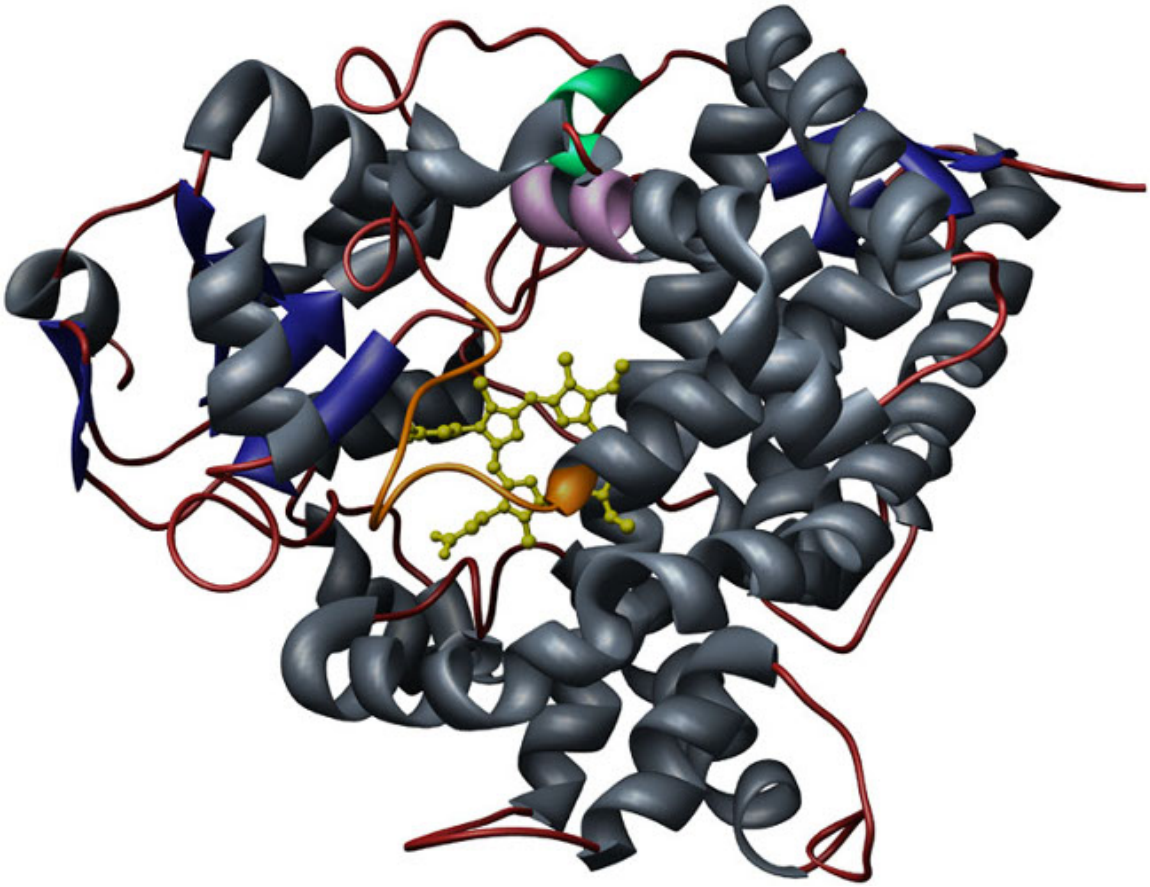


Figure 1.3: Ribbon Diagram of CYP3A4 with Heme Center Highlighted (PDB 1W0E)

Substrate specificities overlap considerably across this superfamily. Because their substrate specificity is broad, CYPs are susceptible to inhibition by a wide variety of

compounds. The CYP enzymes that are involved in the oxidative metabolism of drugs play a major part in the activation and elimination of therapeutic drug molecules. CYP inhibition that leads to decreased elimination and/or changed metabolic pathways of other substrates is a major cause of adverse drug-drug interactions [Lapins et al., 2013]. Drug-drug interactions are therefore an important area of inquiry, especially during the research phase of drug discovery.

1.3 Early Compound Profiling *In Silico*

Drug discovery is a multi-parameter optimization process in which compounds are designed for interaction with their target while simultaneously minimizing off-target activities [Zlokarnik et al., 2005]. In the normal course of taking a compound from hit to lead to drug candidate, adding drug-like properties may minimize the risk of making potent and target-class selective compounds that are biologically inaccessible, but at the same time does little to address the combinatorial complexities of specific drug-drug interactions due to off-target effects.

Because of the complex nature of toxicity, safety prediction is considered more challenging than efficacy prediction [Kruhlak et al., 2012]. Toxicity mechanisms may be unknown or poorly characterized in higher organisms, and similar pathways and targets may be associated with different toxicities and adverse events. Toxicity prediction must also account for many complex interactions and remain alert to the possibility of finding the unexpected. For instance, toxicities could result from on-target effects due to incomplete knowledge or inadequate target validation, or from off-target effects mediated via unknown molecules and mechanisms, or even from genetic variation or comorbidities in any of the previously mentioned pathways.

Techniques for high-throughput *in vitro* screening of CYP inhibition have been developed and implemented broadly in drug discovery pipelines across pharmaceuti-

cal companies and research institutions, resulting in the generation of large, varied sets of data. Some of these data have been released through academic research initiatives (e.g. PubChem AIDs 410 and 1851). These collections enable development of structure-activity relationship models for *in silico* prediction of CYP inhibition by a much larger pool of researchers than those who designed and carried out the assays.

The appeal of *in silico* screening is that, with a steady increase in computing power, screening costs could become negligible. The hope is that virtual compounds could be screened for CYP liabilities to save money and reduce the number of candidates with questionable prospects that would otherwise have to be synthesized [Zlokarnik et al., 2005]. Achieving this would also turn the costly animal toxicity test phase of preclinical research into a validation step rather than a screening step.

Many high-throughput technologies now detect P450 inhibitors. Combined screening data from various assays can guide medicinal chemists away from inhibitory interactions at an early stage. In certain cases it might also identify the inhibition issue allowing intervention by targeted modification of the CYP interacting functionality. According to Zlokarnik, to be generally useful, P450 inhibition screens need to be calibrated against standard methods and preferably also tested with a large set of drugs, for which human drug-drug interaction outcome is known [Zlokarnik et al., 2005]. This should also decrease the number of withdrawals of novel drugs from the market due to inhibition of major P450 isozymes.

The ability to predict clinical safety based on chemical structures has become an increasingly important part of regulatory decision-making. QSAR models are currently used in industry and by regulators to evaluate safety concerns and possible nonclinical effects of a drug when adequate safety data is absent or inconclusive [Kruhlak et al., 2012].

As an example, the United States of America is about to become the last country to accept QSAR in the drug approval process. The draft International Conference on

Harmonisation (ICH) M7 guideline can be viewed as setting a precedent for possible future, broader regulatory applications of QSAR modeling. ICH M7 will for the first time specify that – under very specific conditions – the results of QSAR computational toxicology predictions will be considered sufficient for genotoxic contaminants of pharmaceuticals under consideration and thereby eliminate the need for laboratory testing [Kruhlak et al., 2012].

1.4 Quantitative Structure-Activity Relationship

Quantitative structure-activity relationship modeling is generally accepted as the construction of predictive models of biological activities as a function of structural and molecular information of a compound or compound library [Nantasenamat, 2009].

QSAR models describe the correlation between molecular features and activity at a given end point of interest. Some structure-activity relationship (SAR) models are built from human expert knowledge ("expert rule-based"), but QSAR models are typically defined as those that use mathematical methods to analyze the statistical correlations between molecular features and activity.

A QSAR model that defines the mathematical relationships between descriptors and biological activities of known molecules differs from receptor binding-based efficacy prediction which takes into account binding site characteristics as well as molecular docking analysis. In contrast to QSAR, receptor-binding methods attempt to predict drug efficacy based on known mechanisms of action and medicinal chemistry by individually studying molecular interactions between a drug and targets/receptors [Kruhlak et al., 2012].

The similarity principle holds that untested compounds with molecular features similar to those of known compounds have similar activities and properties. QSAR models therefore predict the biological activity of a compound as a function of its

molecular structure. Published models cover a wide spectrum of biological and physicochemical properties.

Applied QSAR, as described above, has typically been used for drug discovery and development but has also been used to correlate molecular information with other physicochemical properties. This latter approach is termed quantitative structure-property relationship (QSPR). Derived molecular parameters can account for hydrophobicity, topology, electronic properties, and steric effects among other things. These characteristics of compounds can either be determined empirically through experimentation or theoretically via computational chemistry [Nantasenamat, 2009]. The parameters derived from compound structure are referred to as molecular descriptors.

Molecular Descriptors

Molecular descriptors are mathematical representations of the essential information of a molecule in terms of its own physicochemical properties. Depending on the needs of the analysis, properties considered can be electronic, geometric, hydrophobic, constitutional, lipophilic, steric, solubility, quantum chemical, or topological. From a practical viewpoint, molecular descriptors are chemical information that is encoded within the molecular structures [Nantasenamat, 2009].

Molecular features can be either experimentally measured or calculated values. They come in the form of simple physicochemical properties such as logP or logarithmic acid dissociation constant (pKa), numerical representations of substructure fragments, or purely mathematical descriptors. Mathematical descriptors are chemical structural features represented in numerical form, and range from simple atom counts to the product of complex equations that describe electron distribution across a molecule [Kruhlak et al., 2012].

Molecular descriptors as predictors in QSAR modeling are typically less pre-

cise than the "lock and key" relationships that underpin the docking approach to computer-aided drug design. The basic assumption in QSAR modeling is that similar molecules exhibit similar biological activity so that physicochemical properties and/or structural properties of a molecule encoded as molecular descriptors can then be used to predict the biological activity of structurally related compounds with some degree of confidence.

Modeling

QSAR models can be described as global or local. Global models incorporate chemicals with a range of molecular features acting across the spectrum of chemical pathways, whereas local models are highly focused on a single chemical class and end point. Although local models generally have much higher accuracy, their narrow domain of applicability renders them impractical in most regulatory environments where predictions need to be made across a variety of molecules including active pharmaceutical ingredients, as well as metabolites, reagents, and synthetic intermediates [Kruhlak et al., 2012]. Therefore, the focus of this study is global model construction.

The construction of QSAR models typically consists of two main steps:

- Description of molecular structure with derivation of descriptors
- Multivariate analysis correlating molecular descriptors with observed activities.

Intermediate steps, including data preprocessing and statistical evaluation, are also crucial (Figure 1.4) [Nantasenamat, 2009].

A typical QSAR workflow treats chemical structure management, descriptor calculation, and statistical analyses as separate steps that are often performed by non-integrated software packages. Depending on the workflow, this can lower throughput, prevent predictions for new compounds, or prevent model updates when new data

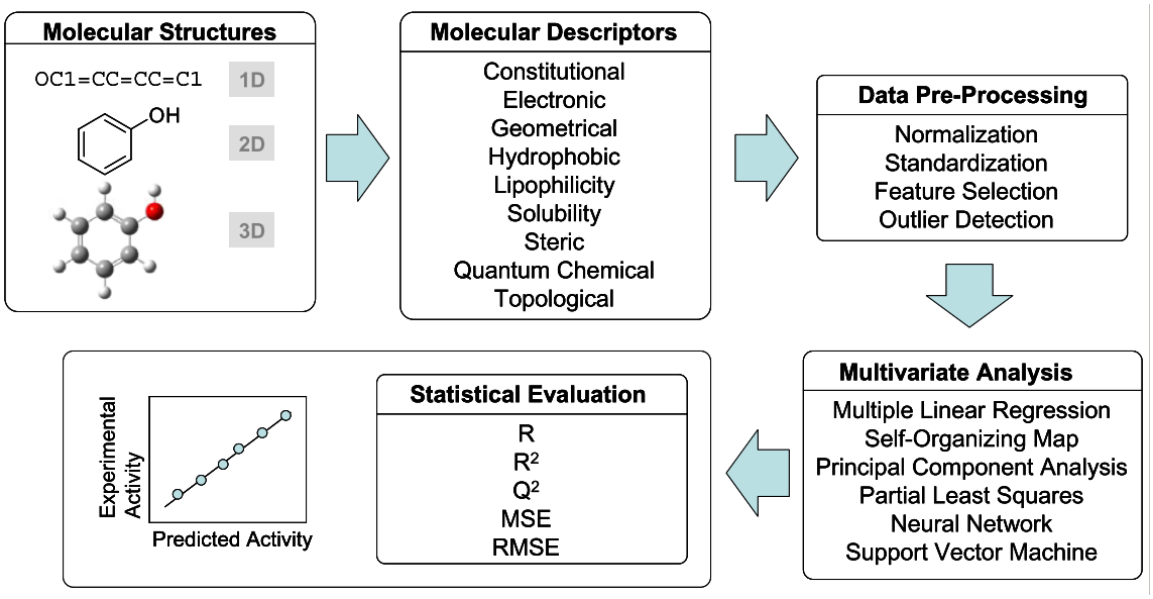


Figure 1.4: Typical QSAR Workflow [Nantasenamat, 2009]

become available. Approaches that integrate as many of these steps as possible are generally preferred.

According to Kruhlak et al., the successful development of a QSAR model for safety prediction requires a sufficient amount of high-quality data, the appropriate selection of descriptors, the availability of one or more suitable statistical or mathematical models and an effective training and validation strategy [Kruhlak et al., 2012].

1.5 Statistical Machine Learning

Machine learning algorithms figure out how to perform important tasks by generalizing from examples. Jan de Leeuw concisely defines statistics as the applied science that constructs and studies techniques for data analysis. Statistical learning refers to a set of approaches for estimating a function that describes a dataset as a precursor for prediction or inference [James et al., 2013]. Statistical machine learning constructs that function by generalizing from examples, that is, from data.

Leo Breiman wrote a landmark paper that documented the beginnings of this

approach. He wrote:

> There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanisms as unknown. [Breiman, 2001]

He argues that the statistical community had traditionally preferred the first view.

Classification is a well understood area of machine learning. A classifier takes a system of inputs, typically a vector of discrete and/or continuous feature values and then outputs a single discrete value, the class [Domingos, 2012]. Given labelled examples, a classifier assigns a new item to a class based on its characteristics.

The current state of machine learning is fundamentally a subset of optimization and has found its biggest successes in fields where there are far more variables than parameters. The ultimate goal of machine learning in these cases is to generalize beyond examples in a training set, because no matter how many examples the training set contains, it is unlikely that those exact examples will be seen again in practical applications [Domingos, 2012].

In the context of QSAR then, with enough prior knowledge and sound assay results, machine learning may be a practical approach to fill in the gaps of clinical knowledge for any relevant CYP isozyme when queried against any untested compound of interest so long as the compound structure is known.

This is a wider view than traditionally encountered in most applied science education where simple hypothesis testing predominates. Opening up the field of drug discovery to data analysis like this brings opportunity for exploration but also new concerns, such as the 'curse of dimensionality', 'degrees of freedom of the analyst', 'black box algorithms', bias estimation difficulties and the 'no free lunch theorem' [Boulesteix and Schmid, 2014].

1.6 Sources of Data for Learning

Systems biology research encompasses the generation of high-throughput datasets of system components (omics data), experimental methods of analysis and data integration, as well as the development and application of network approaches and computationally derived models. In pharmaceutical research, systems biology efforts are directed towards the identification of drug targets, the development of novel therapeutics and new indications for existing drugs [Berg, 2014].

Omics tools, developed over the past several decades, can provide global information on the levels and dynamic changes in cellular and tissue components at specific time points in samples from cell-based assays, preclinical animal models and human studies. Omics data sets derived from transcriptomics, proteomics, and metabolomics can also be used and integrated with each other, as well as with genomics information and other data types, to construct models of cell signaling, pathway and disease networks. These models may help to identify new targets as well as better understand and predict drug action *in vivo*.

> The ultimate goal of systems biology in this context is an understanding of physiology and disease across the multiple hierarchical levels of organization, from chemical and molecular interactions to pathways and pathway networks, at the cell-cell and tissue level, organs and organ systems and, ultimately, to the functioning of the whole organism [Berg, 2014].

The data available for learning in discovery and development of pharmaceuticals tends to be compound-centric. Compound data collections are generally concerned with the identification and characterization of small molecules or biologics that selectively inhibit (or activate) specific molecular targets or pathway mechanisms. These kinds of studies are particularly useful to drug discovery research as they attempt

to work out related drug mechanisms of action. These studies also inform secondary drug development goals, such as clinical indication selection and patient stratification [Berg, 2014].

1.7 Reproducibility

Several recent publications have highlighted the negative impact of irreproducible biomedical research, such as the group from Bayer that claimed they had to halt two-thirds of their research efforts in 2011 [Prinz et al., 2011] and the group from Amgen that reported only an 11% success rate in trying to replicate the effects of major cancer drug findings [Begley and Lee, 2012]. In the Amgen case, the findings tested were not disclosed. In the biological sciences, data generation outpaces analysis and knowledge creation.

Some of the issues to overcome in this area include the need for more biologists trained in quantitative and statistical methods for analyzing large data sets and the open release of findings and experimental protocols.

Open science confers the following benefits:

- Reproducibility of experimentation allows other researchers to use the exact methods to calculate the relations between biological data.

- Faster development of disease models and therapeutic treatments due to the reuse of existing knowledge. Projects can be built upon existing results more easily or used to extend the research in directions unanticipated by the original team. First-pass results can be subject to new analysis and a second look at compounds with interesting side effects can lead to serendipitous discoveries.

- Increased quality as a result of having more researchers studying the same topic to provide a layer of assurance that errors will not propagate.

- Long-term availability of data and code. If these resources are not tied to businesses or patents, then they can be posted to multiple repositories to ensure that they are available in the future [Prlić and Lapp, 2012].

New research findings, supporting data and methods should therefore be made public for independent verification and replication, so as not to delay medical advances.

1.8 Aims

This project builds QSAR models that classify compounds, from structure alone, as inhibitors or non-inhibitors of five cytochrome P450 isozymes, as a way to flag the risk of off-target effects in candidate molecules.

I compare a well-accepted commercial method of binary classification with three open-source machine learning methods, according to the following plan:

- Build Binary QSAR models in the Molecular Operating Environment (MOE).
- Develop and implement comparable methods in open source software.
- Evaluate and compare results from all models.
- Perform this analysis as reproducible research.

CHAPTER 2

Materials and Methods

2.1 Review of PubChem AID 1851

PubChem AID 1851 contains data for inhibition of five major CYP isozymes (1A2, 2C9, 2C19, 2D6 and 3A4) by 17,143 chemical compounds [Veith et al., 2009]. The tested compounds were all drugs or drug-like compounds. Most of these compounds had molecular weight below 500 daltons and logP below 5 [Lapins et al., 2013].

PubChem AID 1851 used low-luminescent substrates, which are converted to more-luminescent metabolites as the signal for this assay. Reaction progress was measured as increasing light intensity during CYP metabolism of the substrate. Inhibitors of a particular CYP isozyme reduce the rate of substrate metabolism and thus decrease the luminescent signal [Zlokarnik et al., 2005].

The most recent CYP inhibition assay technology is based on substrates that release luciferin as the metabolite. This is a coupled assay system in which addition of luciferase and ATP converts the freed luciferin to des-carboxyluciferin along with light emission. The format is similar to fluorescence methods, but particularly well-suited to high-throughput applications because all that is required are addition-only manipulations and luminescence plate readers [Zlokarnik et al., 2005]. The assay obtained

luminescence readings at a range of compound concentrations and then determined activity parameters using the Hill equation.

In PubChem AID 1851, compounds are classified as inhibitors or non-inhibitors for each CYP with an activity cutoff set to AC50 = 10 μM (AC50, "activity concentration 50", refers to the concentration that is required to elicit half-maximal effect). However, in cases where the dose-response curve for a compound showed poor fit or the inhibition efficacy was below 60%, the assay results were regarded as inconclusive [Veith et al., 2009].

Each compound received an Activity Score from 0 to 100, which combines its AC50 with confidence measures for dose-response curve completeness and inhibition efficacy. Higher scores indicate stronger inhibition, higher confidence, or both. Compounds scoring 40–100 are inhibitors, those scoring 0 are non-inhibitors, and those scoring 1–39 are inconclusive [Lapins et al., 2013]. In this study, inconclusive compounds were not discarded: every compound with an Activity Score below 40 was labelled a non-inhibitor for binary classification.

The data from PubChem AID 1851 are available for download from the NIH through the PubChem website. I downloaded two files comprising the entire data set. The first, a structures file, contained the structural information encoded in the simplified molecular-input line-entry system (SMILES) format for each tested compound with corresponding Structure ID (SID) and Compound ID (CID) as assigned by NCBI. The second, also organized by SID and CID, contained all luminescent responses from the high-throughput screen and the fitted parameters summarized by the Activity Score.

2.2 Data Set Preparation and Molecular Descriptor Generation

Both data files from PubChem AID 1851 were downloaded as comma separated value (.csv) files and merged on the SID column using Python's pandas library, a library for handling, manipulating and reshaping structured data.

Preprocessing typically begins with data cleaning, because raw data often contain anomalies, errors, or inconsistencies such as missing data, incomplete data, and invalid character values which may cause trouble in data analysis if left untreated. It is more complicated when data are collated from many formats that require harmonization and redundancy elimination [Nantasenamat, 2009]. Here, the source files needed little preprocessing.

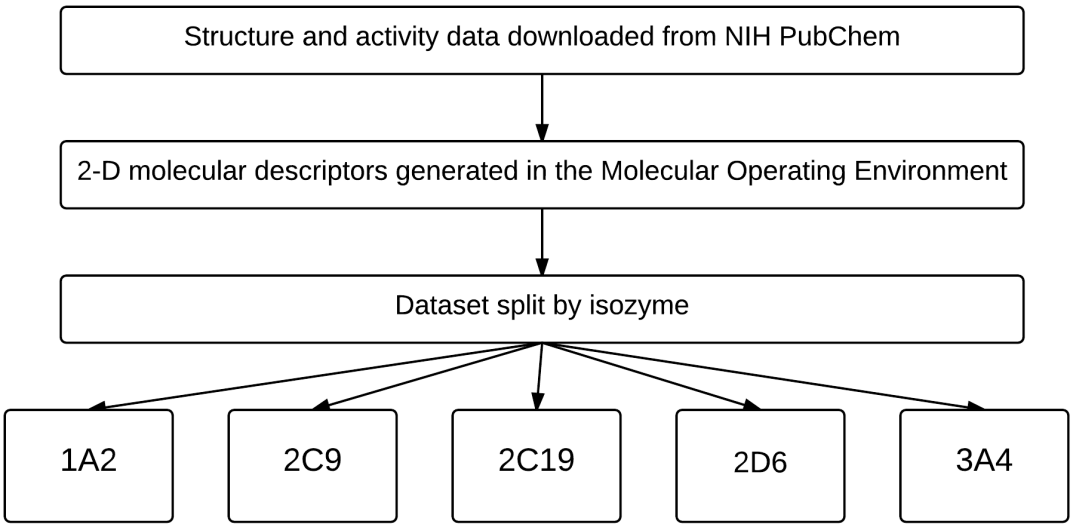


Figure 2.1: Data Download and Preparation

Molecular Descriptor Generation

The merged file was loaded into a database in the Molecular Operating Environment (MOE). MOE's wash function removed salts and generated an energy-minimized conformation of each compound. Calculating the full suite of MOE 2-D descriptors added 186 columns of nominal, ordinal and continuous values to the database. The resulting master file was saved in .csv format.

The dataset was then split by isozyme into five separate files that each contained only the SID, the Activity Score, and the 186 MOE 2-D descriptors using a script in the Python programming language.

For each isozyme, non-inhibitors far outnumbered inhibitors. Because some later analyses require balanced classes, a Python script randomly shuffled each isozyme's non-inhibitors and kept only as many as there were inhibitors.

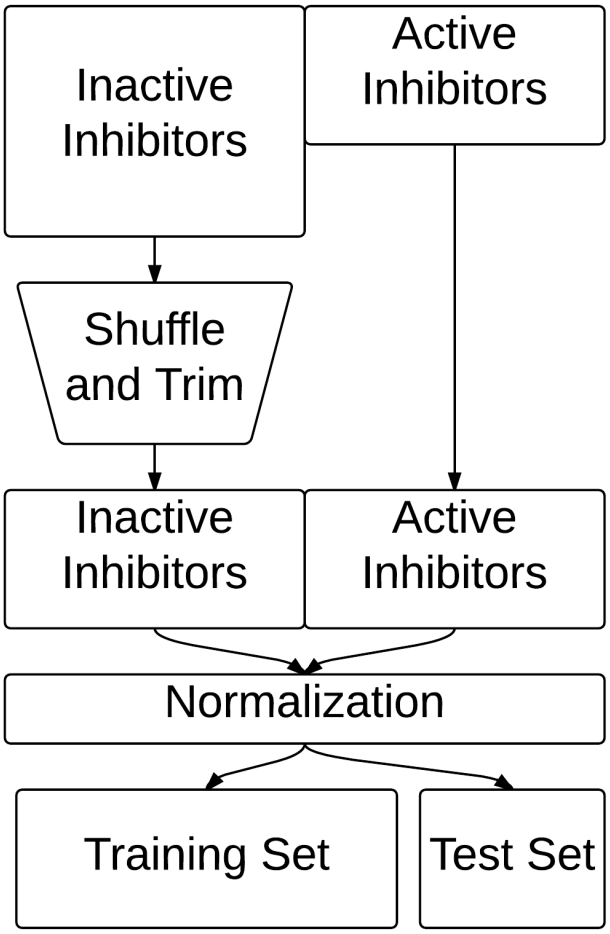


Figure 2.2: Isozyme Data Set Preparation

Next the script randomly shuffled the balanced datasets and split them into a training set and a test set. The training set held 80% of compounds and the test set 20%. The split was not stratified by class; the class ratio in each split was inspected to confirm it remained balanced.

For each use of randomness in computation, a seed number was set for the random number generator to ensure reproducible results.

The balanced and split data sets are archived on Figshare for permanent, free and open access (http://dx.doi.org/10.6084/m9.figshare.1181846) and (http://dx.doi.org/10.6084/m9.figshare.1066108). All subsequent analyses use these

same splits for comparability.

2.3 Data Preprocessing

Data sets often contain redundant or noisy variables that make it harder for learning algorithms to discern meaningful patterns. Such multicollinearity can be exploited, or reduced to cut the computational cost of model construction [Nantasenamat, 2009].

The variables in this data set varied widely in range and distribution, which causes problems for algorithms that use distance measures. Min-max normalization and z-score standardization address this.

In min-max normalization, the minimum and maximum value of each variable is adjusted to a uniform range between 0 and 1 according to the following equation: $X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$ In z-score standardization, each variable is centered to mean 0 and scaled to unit variance: $z = \frac{x - \mu}{\sigma}$ [Nantasenamat, 2009]

When data do not have a Gaussian (normal) distribution, simple transformations can make them more normal or symmetric. A common approach is a logarithmic transformation. This is typically performed on dependent variables such as the modeled biological/chemical properties of interest whereby IC50 may be transformed to logIC50 or -logIC50. [Nantasenamat, 2009]

2.4 Modeling - Mapping Descriptors to Activity Data

At this point, each isozyme had a complete data set, balanced between inhibitors and non-inhibitors and split into a training set and a test set. The preceding sections described data balancing and normalization; the following sections outline classification algorithms and validation procedures used in this study.

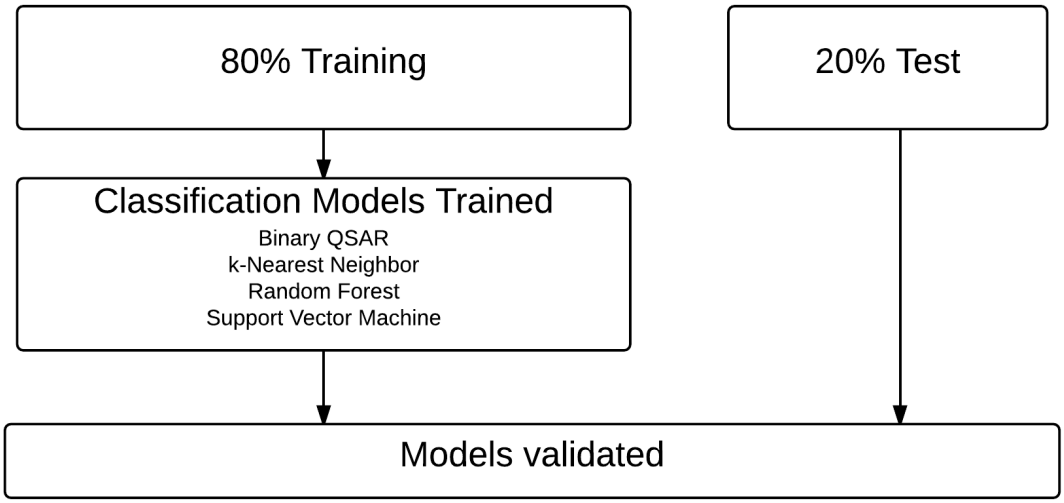


Figure 2.3: Supervised Classification Model Training and Validation

2.4.1 Binary QSAR in the Molecular Operating Environment

The Binary QSAR approach described by Labute [Labute, 1999] and as implemented in the 2011 version of MOE, classifies activity with a Bayesian model after reducing the descriptors to principal components. Principal component analysis (PCA) projects the data onto a smaller set of axes that point in the directions of greatest variance. The first principal component explains the most variance, the second the next most, and so on. Models are subsequently built with the original data projected onto any or all of these principal components.

Earlier work in Dr. Zheng's lab found that partial least squares (PLS) regression, a quantitative method in MOE that also begins with PCA, predicted poorly on this data set.

The training data were loaded into MOE and Binary QSAR was run from its menu-driven interface. The Activity Score threshold was set to 39, so compounds scoring 40 or above were labelled inhibitors and those scoring 39 or below non-inhibitors.

The smoothing parameter was left at the default value of 0.25. MOE automatically performs principal component analysis on high-dimensional datasets before Binary QSAR; it offers no way to use the full suite of molecular descriptors untransformed.

For each isozyme, several models were built, each with a different number of principal components. Principal components are mutually orthogonal, so each captures a separate portion of the total variance. The assumption is that inclusion of more principal components leads to more of the variance being accounted for in a classification decision. However, since each principal component is a linear combination of all the variables, the benefits of dimensionality reduction come at the cost of interpretability. One aim of this study is to find the fewest principal components that still give good predictions. For exploratory purposes, models with 2, 5, 10, 15, 20, 30, and 44 principal components were constructed.

MOE models were written to .fit files and the model report saved as a .txt file.

Modeling in Python

Python is a dynamically typed, object-oriented, interpreted language. Its main strengths are rapid prototyping and a mature ecosystem of libraries, such as scikit-learn for machine learning. It is easy to learn, with abundant free learning materials online. The following classifiers were implemented with scikit-learn.

2.4.2 *k*-Nearest Neighbors (*k*-NN)

The k-NN algorithm predicts the class of an object from the classes of its k most similar training set objects [Lapins et al., 2013]. It finds the k nearest points under a distance measure such as Euclidean distance and assigns the majority class among them.

2.4.3 Random Forest (RF)

A random forest is an ensemble of decision trees. Decision trees are made of nodes and branches. At each node, the data set is split based on the value of some attribute that is selected so that the instances of different classes are predominately moved to different branches. Classification passes each instance from the root node along the branches to a leaf node. To introduce diversity between the trees of a random forest, a small random subset of features is considered at each node of each tree. The forest classifies by majority vote of all trees [Lapins et al., 2013].

2.4.4 Support Vector Machines (SVM)

An SVM finds the hyperplane that separates the classes with the greatest margin. SVMs use a kernel function to project the data into a higher-dimensional feature space before finding the hyperplane [Lapins et al., 2013]. The margin is the minimum distance from the data points to the hyperplane. The points closest to the hyperplane, called support vectors, determine its position.

2.5 Model Evaluation and Validation

The building blocks of a successful QSAR model are the accuracy of the input data, selection of appropriate descriptors and statistical tools, and most importantly validation of the developed model. Validation is the process where reliability and relevance of a procedure are established for a predetermined purpose. For QSAR models validation should target robustness, prediction performance and applicability domain of the models [Lapins et al., 2013].

Statistical evaluation in QSAR modeling is essential to validate the model as well as to evaluate its predictive performance. Predictive performance can be assessed by

dividing the data into a training set, used to construct the model, and a test set, used to evaluate it. Internal performance is typically assessed from the predictive performance within a training set, while external performance of a classifier can be assessed from the predictive performance against an independent test set that has never been seen by the training model.

This study used two kinds of metric: overall accuracy, and the true positive and true negative rates. A common approach to model assessment in the exploratory phase is N-fold cross-validation: the data set is partitioned into N folds, and each fold in turn is left out of model construction and used for validation. For example, 5-fold cross-validation trains each model on four folds and validates it on the fifth. When samples are limited, leave-one-out cross-validation is the preferred approach. In that case, the number of folds is equal to the number of samples present in the data set [Nantasenamat, 2009]. Using these methods with the original training set allows the original test set to be preserved for final validation.

In this study, all Python models were assessed by 5-fold cross-validation on the training set. Each model was then fit on the full training set and used to predict the test set, which played no part in model fitting or feature scaling. The MOE models were built on the same training set and evaluated on the same test set, but without cross-validation.

Confusion Matrix

A confusion matrix (also called an error matrix) summarizes a classifier's performance. Each data point is counted by its actual class along one axis and by its predicted class along the other. For binary classification, this gives a 2×2 matrix.

		Prediction Class	
		p	**n**
Actual Class	p′	True Positive	False Negative
	n′	False Positive	True Negative

In this study, actual inhibitors (according to the assay) that the model predicts as inhibitors are true positives (TP), and actual non-inhibitors predicted as non-inhibitors are true negatives (TN). Actual inhibitors predicted as non-inhibitors are false negatives (FN), or Type II errors. Non-inhibitors predicted as inhibitors are false positives (FP), or Type I errors.

Accuracy

Accuracy is the fraction of instances classified correctly:

$$ACC = \frac{(TP + TN)}{(TP + FP + TN + FN)}$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives or over-predictions, and FN is the number of false negatives or missed predictions.

Accuracy is not an optimal measure of model performance if the data set is unbalanced (i.e. sizes of the classes are unequal) or if some errors are more costly than others (e.g. false negatives compared to false positives) [Lapins et al., 2013]. For this reason, all data sets were balanced to contain roughly equal numbers of inhibitors and non-inhibitors.

I also calculated the true positive rate (TPR) and true negative rate (TNR) for

all models. TPR is the fraction of actual inhibitors predicted correctly:

$$\text{TPR} = \frac{TP}{TP + FN}$$

TPR is also called sensitivity; MOE calls it Accuracy on Actives.

TNR is the fraction of actual non-inhibitors predicted correctly:

$$\text{TNR} = \frac{TN}{TN + FP}$$

TNR is also called specificity; MOE calls it Accuracy on Inactives.

Binary QSAR Test Set Evaluation

The test sets were loaded and the washed structures were appended to the .csv files. All models were evaluated using MOE's menu-driven workflow and the classification probabilities were appended to the database file and saved as a .csv.

In Microsoft Excel, predicted probabilities ≥ 0.5 were classified as inhibitors and those < 0.5 as non-inhibitors. Confusion matrices, accuracy, TPR and TNR were then tabulated in the spreadsheet.

Python Classifiers Using the Full Set of 2-D Descriptors

The Python models omit PCA and use all 186 descriptors. For each isozyme, the Activity Score was converted to a binary response (1 for scores ≥ 40, 0 otherwise), and all 186 descriptors served as predictors. Each classifier was combined with a standardization step (mean 0, standard deviation 1) in a scikit-learn pipeline, so the scaling parameters were estimated only from the data each model was fit on. Hyperparameters matched the scikit-learn 0.15 defaults: k-NN with $k = 5$; a random forest of 10 trees considering $\sqrt{186}$ features at each split; and an SVM with a radial basis function kernel, $C = 1$ and $\gamma = 1/186$. Each model was first scored by 5-

fold cross-validation on the training set, then fit on the full training set and used to predict the held-out test set. Accuracy, TPR and TNR were calculated from the resulting confusion matrices, and the area under the receiver operating characteristic curve (AUC) from each model's predicted probability (or, for the SVM, its decision function value).

One notebook per isozyme runs the three classifiers through a shared evaluation module. Code and results are documented in Jupyter notebooks hosted on GitHub (https://github.com/5x5x5x5/CYP/), so the analysis is easy to verify and extend. Because scikit-learn gives every classifier the same interface, trying another algorithm requires only adding it to the list of classifiers.

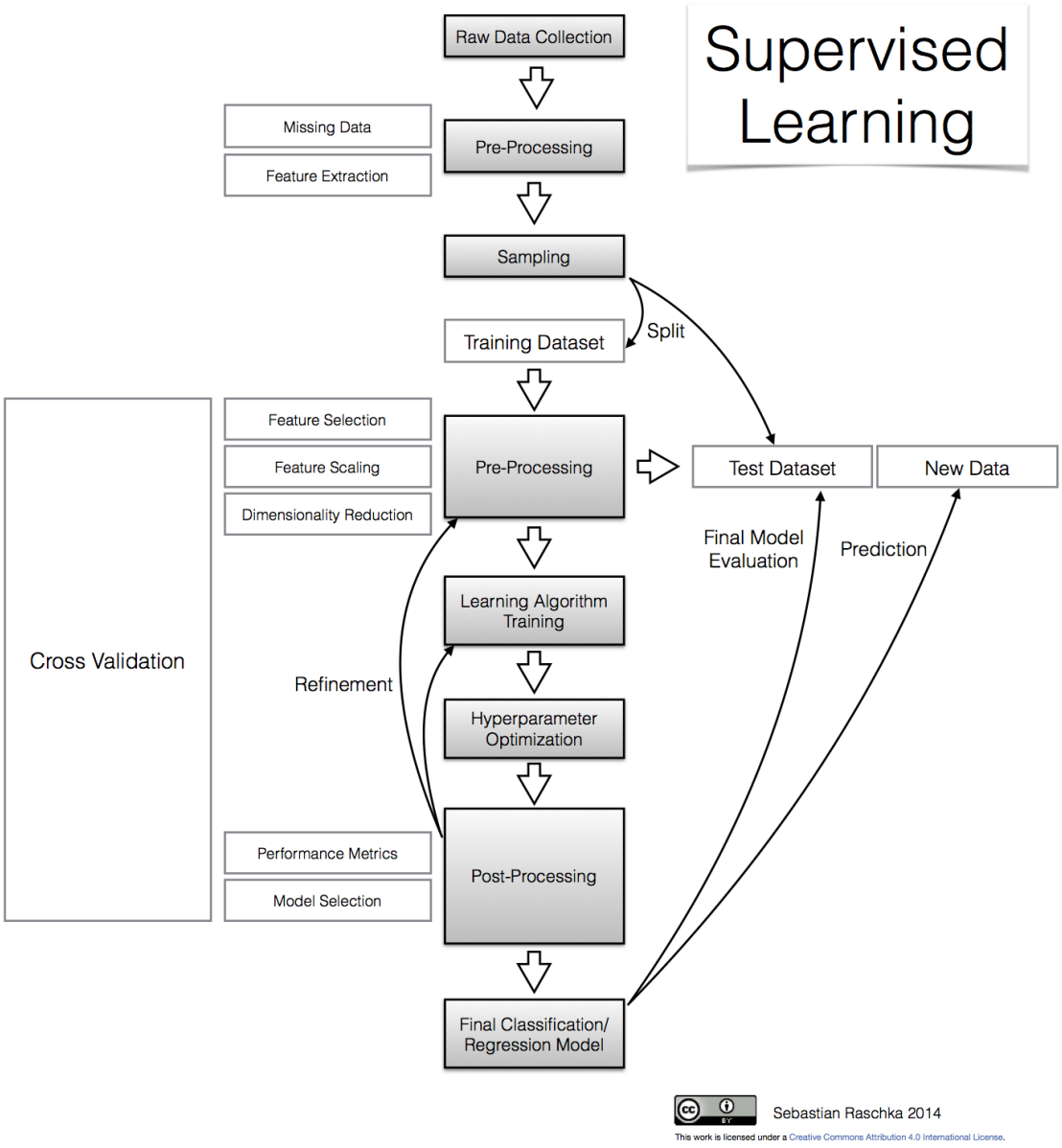


Figure 2.4: Supervised Learning Workflow for Classification

CHAPTER 3

Results

This chapter first reports results for MOE Binary QSAR models built with 2 to 44 principal components, then compares the best MOE configuration with the three Python classifiers for each isozyme. All accuracies are the fraction of compounds classified correctly. I treat a model as good if its total accuracy exceeds 0.6.

Three patterns hold for the MOE models on every isozyme. First, accuracy on inhibitors is higher than accuracy on non-inhibitors, with total accuracy in between. Second, training and test set accuracies are close, suggesting little overfitting. Third, about 20 principal components maximizes total accuracy while avoiding the drop in accuracy on non-inhibitors seen at both ends of the range. The method comparisons therefore use the 20-component MOE model. For the Python classifiers, training set accuracies are 5-fold cross-validation estimates; test set accuracies come from models fit on the full training set and then applied to the test set.

3.1 1A2 MOE Models

Table 3.1: 1A2 MOE Model Results

1A2 Training Set Accuracy			
PC	Total	Active	Inactive
2	0.638	0.752	0.522
5	0.704	0.746	0.662
10	0.734	0.755	0.712
15	0.737	0.770	0.704
20	0.741	0.781	0.701
30	0.735	0.782	0.686
44	0.725	0.777	0.672

1A2 Test Set Accuracy			
PC	Total	Active	Inactive
2	0.632	0.758	0.517
5	0.705	0.773	0.643
10	0.746	0.749	0.743
15	0.752	0.775	0.731
20	0.748	0.779	0.721
30	0.739	0.783	0.698
44	0.720	0.773	0.672

The 1A2 models were trained on 4820 inhibitors and 4780 non-inhibitors and tested on 1180 inhibitors and 1220 non-inhibitors.

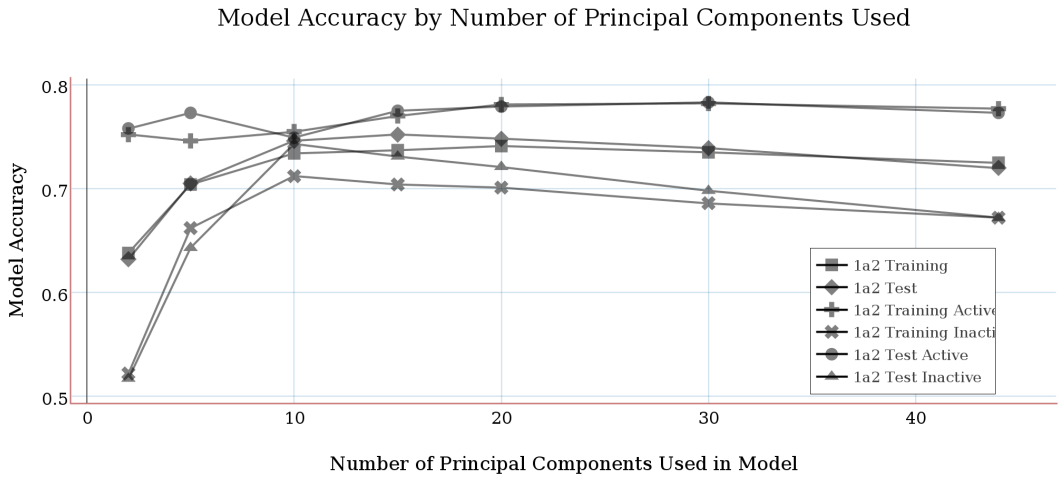


Figure 3.1: 1A2 MOE Model Accuracy

With 20 principal components (Figure 3.1), test set accuracy is 0.748 overall, 0.779 on inhibitors and 0.721 on non-inhibitors.

3.2 2C9 MOE Models

Table 3.2: 2C9 MOE Model Results

2C9 Training Set Accuracy			
PC	Total	Active	Inactive
2	0.620	0.695	0.547
5	0.683	0.722	0.645
10	0.699	0.744	0.655
15	0.702	0.752	0.653
20	0.710	0.756	0.665
30	0.711	0.754	0.668
44	0.712	0.770	0.655

2C9 Test Set Accuracy			
PC	Total	Active	Inactive
2	0.609	0.702	0.510
5	0.675	0.713	0.634
10	0.682	0.730	0.632
15	0.686	0.739	0.630
20	0.685	0.731	0.636
30	0.690	0.730	0.646
44	0.688	0.750	0.621

The 2C9 models were trained on 3266 inhibitors and 3327 non-inhibitors and tested on 855 inhibitors and 794 non-inhibitors.

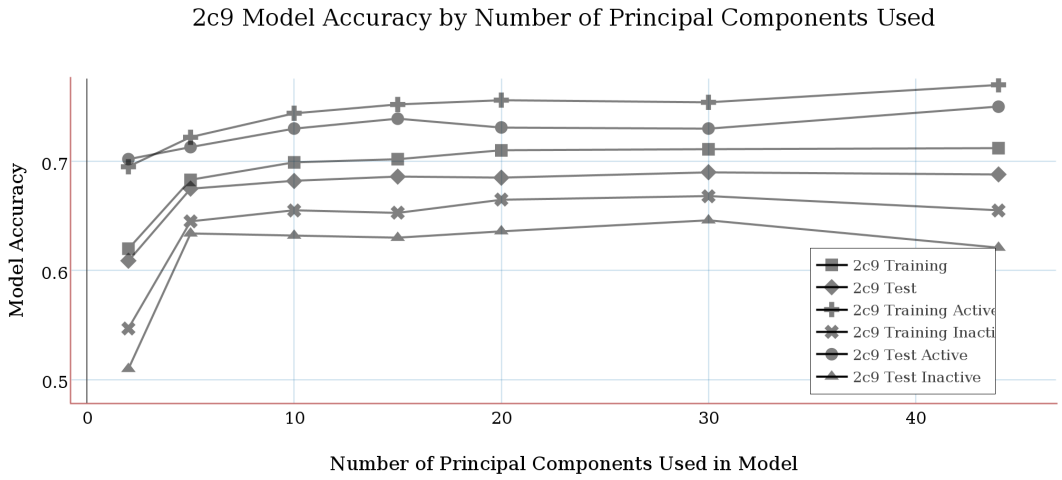


Figure 3.2: 2C9 MOE Model Accuracy

With 20 principal components (Figure 3.2), test set accuracy is 0.685 overall, 0.731 on inhibitors and 0.636 on non-inhibitors.

3.3 2C19 MOE Models

Table 3.3: 2C19 MOE Model Results

2C19 Training Set Accuracy			
PC	Total	Active	Inactive
2	0.585	0.774	0.396
5	0.685	0.726	0.645
10	0.699	0.736	0.661
15	0.700	0.747	0.653
20	0.705	0.749	0.661
30	0.717	0.761	0.672
44	0.708	0.777	0.640

2C19 Test Set Accuracy			
PC	Total	Active	Inactive
2	0.593	0.775	0.408
5	0.683	0.729	0.635
10	0.691	0.731	0.650
15	0.687	0.736	0.637
20	0.698	0.758	0.638
30	0.699	0.764	0.633
44	0.694	0.774	0.613

The 2C19 models were trained on 4721 inhibitors and 4741 non-inhibitors and tested on 1193 inhibitors and 1173 non-inhibitors.

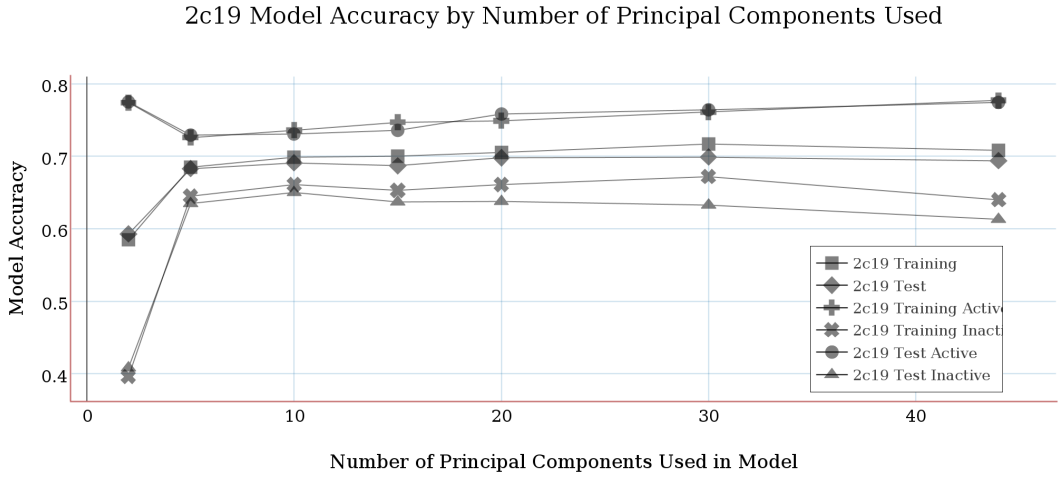


Figure 3.3: 2C19 MOE Model Accuracy

With 20 principal components (Figure 3.3), test set accuracy is 0.698 overall, 0.758 on inhibitors and 0.638 on non-inhibitors.

3.4 2D6 MOE Models

Table 3.4: 2D6 MOE Model Results

2D6 Training Set Accuracy			
PC	Total	Active	Inactive
2	0.589	0.753	0.426
5	0.670	0.706	0.634
10	0.685	0.721	0.648
15	0.686	0.709	0.662
20	0.705	0.721	0.689
30	0.703	0.732	0.673
44	0.690	0.725	0.656

2D6 Test Set Accuracy			
PC	Total	Active	Inactive
2	0.590	0.738	0.443
5	0.667	0.685	0.650
10	0.662	0.688	0.636
15	0.665	0.692	0.639
20	0.683	0.699	0.668
30	0.686	0.696	0.677
44	0.669	0.707	0.632

The 2D6 models were trained on 2219 inhibitors and 2214 non-inhibitors and tested on 552 inhibitors and 557 non-inhibitors.

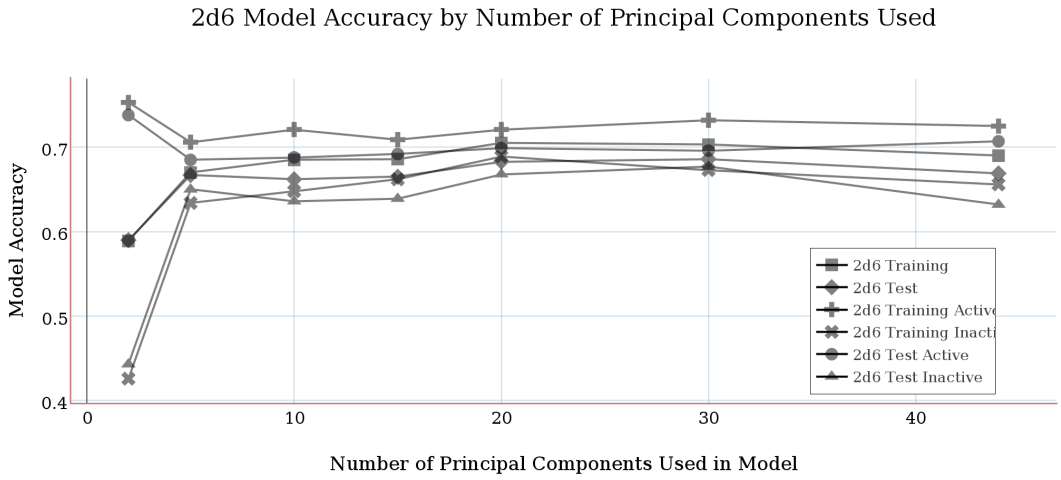


Figure 3.4: 2D6 MOE Model Accuracy

With 20 principal components (Figure 3.4), test set accuracy is 0.683 overall, 0.699 on inhibitors and 0.668 on non-inhibitors.

3.5 3A4 MOE Models

Table 3.5: 3A4 MOE Model Results

3A4 Training Set Accuracy			
PC	Total	Active	Inactive
2	0.644	0.699	0.589
5	0.675	0.727	0.623
10	0.698	0.741	0.655
15	0.701	0.746	0.656
20	0.705	0.749	0.660
30	0.709	0.761	0.657
44	0.698	0.773	0.623

3A4 Test Set Accuracy			
PC	Total	Active	Inactive
2	0.627	0.683	0.573
5	0.656	0.719	0.595
10	0.677	0.719	0.637
15	0.680	0.733	0.628
20	0.686	0.736	0.638
30	0.686	0.742	0.631
44	0.676	0.747	0.607

The 3A4 models were trained on 4234 inhibitors and 4193 non-inhibitors and tested on 1033 inhibitors and 1074 non-inhibitors.

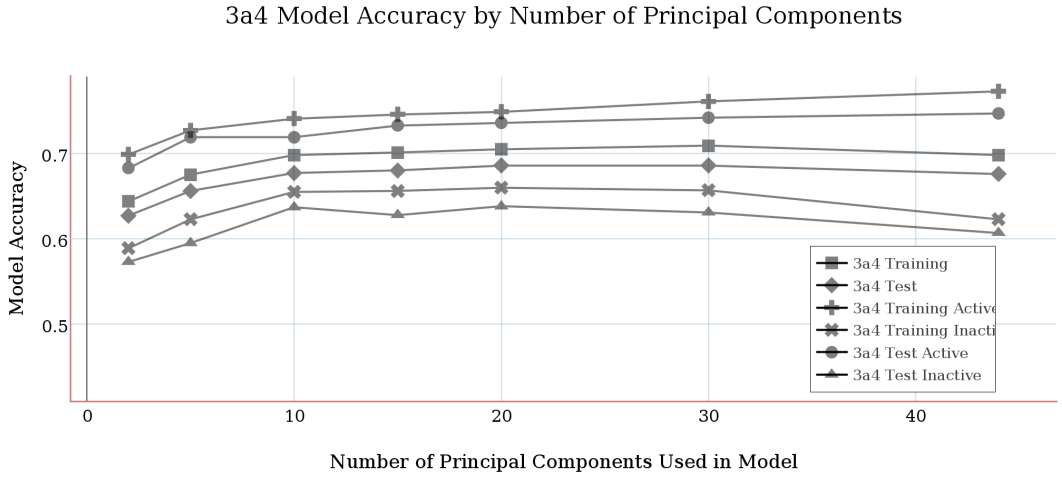


Figure 3.5: 3A4 MOE Model Accuracy

With 20 principal components (Figure 3.5), test set accuracy is 0.686 overall, 0.736 on inhibitors and 0.638 on non-inhibitors.

3.6 1A2 Classification Method Comparison

Table 3.6: Comparison of Classification Methods for 1A2. Python training accuracies are 5-fold cross-validation estimates.

1A2 Classification Method Comparison		
	Training Set	Test Set
MOE 20PC	0.741	0.748
k-NN	0.765	0.780
Random Forest	0.775	0.792
SVM	0.798	0.808

On the 1A2 test set, MOE Binary QSAR with 20 principal components reached an accuracy of 0.748, compared with 0.780 for k-NN, 0.792 for random forest and 0.808 for SVM.

3.7 2C9 Classification Method Comparison

Table 3.7: Comparison of Classification Methods for 2C9. Python training accuracies are 5-fold cross-validation estimates.

2C9 Classification Method Comparison		
	Training Set	Test Set
MOE 20PC	0.710	0.685
k-NN	0.719	0.712
Random Forest	0.725	0.725
SVM	0.748	0.754

On the 2C9 test set, MOE Binary QSAR with 20 principal components reached an accuracy of 0.685, compared with 0.712 for k-NN, 0.725 for random forest and 0.754 for SVM.

3.8 2C19 Classification Method Comparison

Table 3.8: Comparison of Classification Methods for 2C19. Python training accuracies are 5-fold cross-validation estimates.

2C19 Classification Method Comparison		
	Training Set	Test Set
MOE 20PC	0.705	0.698
k-NN	0.731	0.746
Random Forest	0.734	0.733
SVM	0.764	0.780

On the 2C19 test set, MOE Binary QSAR with 20 principal components reached an accuracy of 0.698, compared with 0.746 for k-NN, 0.733 for random forest and 0.780 for SVM.

3.9 2D6 Classification Method Comparison

Table 3.9: Comparison of Classification Methods for 2D6. Python training accuracies are 5-fold cross-validation estimates.

2D6 Classification Method Comparison		
	Training Set	Test Set
MOE 20PC	0.705	0.683
k-NN	0.717	0.718
Random Forest	0.728	0.729
SVM	0.759	0.752

On the 2D6 test set, MOE Binary QSAR with 20 principal components reached an accuracy of 0.683, compared with 0.718 for k-NN, 0.729 for random forest and 0.752 for SVM.

3.10 3A4 Classification Method Comparison

Table 3.10: Comparison of Classification Methods for 3A4. Python training accuracies are 5-fold cross-validation estimates.

3A4 Classification Method Comparison		
	Training Set	Test Set
MOE 20PC	0.705	0.686
k-NN	0.714	0.704
Random Forest	0.718	0.728
SVM	0.754	0.746

On the 3A4 test set, MOE Binary QSAR with 20 principal components reached an accuracy of 0.686, compared with 0.704 for k-NN, 0.728 for random forest and 0.746 for SVM.

3.11 Python Classifier Test Set Results

Table 3.11: Python Classifier Test Set Results

Isozyme	Method	Accuracy	TPR	TNR	AUC
1A2	k-NN	0.780	0.825	0.737	0.852
	Random Forest	0.792	0.775	0.808	0.866
	SVM	0.808	0.849	0.768	0.886
2C9	k-NN	0.712	0.749	0.673	0.771
	Random Forest	0.725	0.705	0.746	0.791
	SVM	0.754	0.806	0.698	0.816
2C19	k-NN	0.746	0.786	0.705	0.803
	Random Forest	0.733	0.724	0.742	0.807
	SVM	0.780	0.837	0.723	0.845
2D6	k-NN	0.718	0.746	0.689	0.780
	Random Forest	0.729	0.649	0.808	0.795
	SVM	0.752	0.726	0.777	0.830
3A4	k-NN	0.704	0.749	0.660	0.771
	Random Forest	0.728	0.694	0.761	0.815
	SVM	0.746	0.802	0.692	0.831

TPR is the fraction of actual inhibitors predicted correctly, TNR the fraction of actual non-inhibitors predicted correctly, and AUC the area under the receiver operating characteristic curve. k-NN and SVM were more accurate on inhibitors than on non-inhibitors for most isozymes, whereas random forest was consistently more accurate on non-inhibitors.

Figure 3.6 shows the receiver operating characteristic (ROC) curves for the three Python classifiers on each test set. The curves lie well above the chance diagonal, and the SVM curve lies at or above the other two across most of the range for every isozyme.

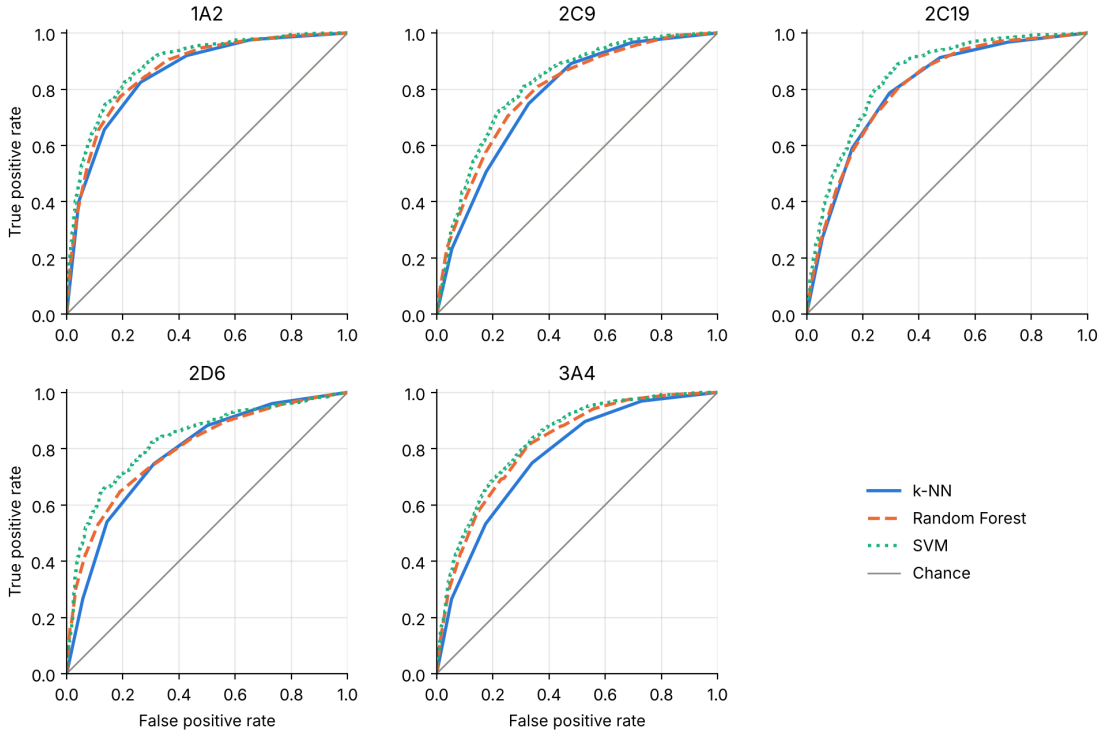


Figure 3.6: Receiver operating characteristic curves on the test set for each isozyme. Each classifier was fit on the training set; the diagonal marks chance performance.

3.12 Overall Method Comparison

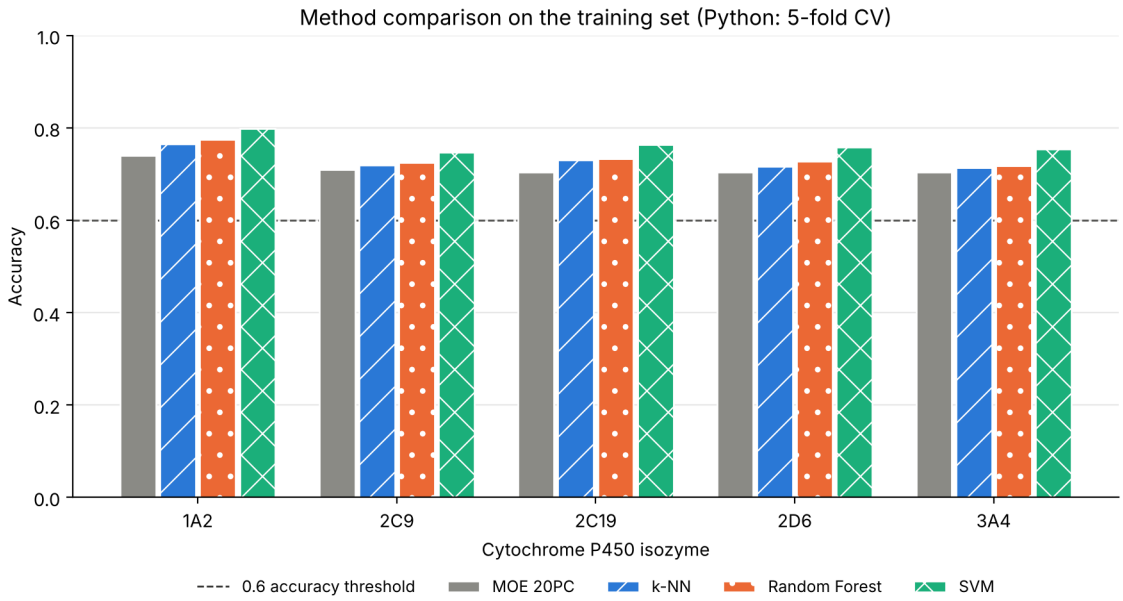


Figure 3.7: Inhibitor Classification Method Comparison on the Training Set

On the training set (cross-validated for the Python classifiers), all three Python classifiers matched or exceeded MOE Binary QSAR for every isozyme, and the support vector machine classifier had the highest accuracy in each case.

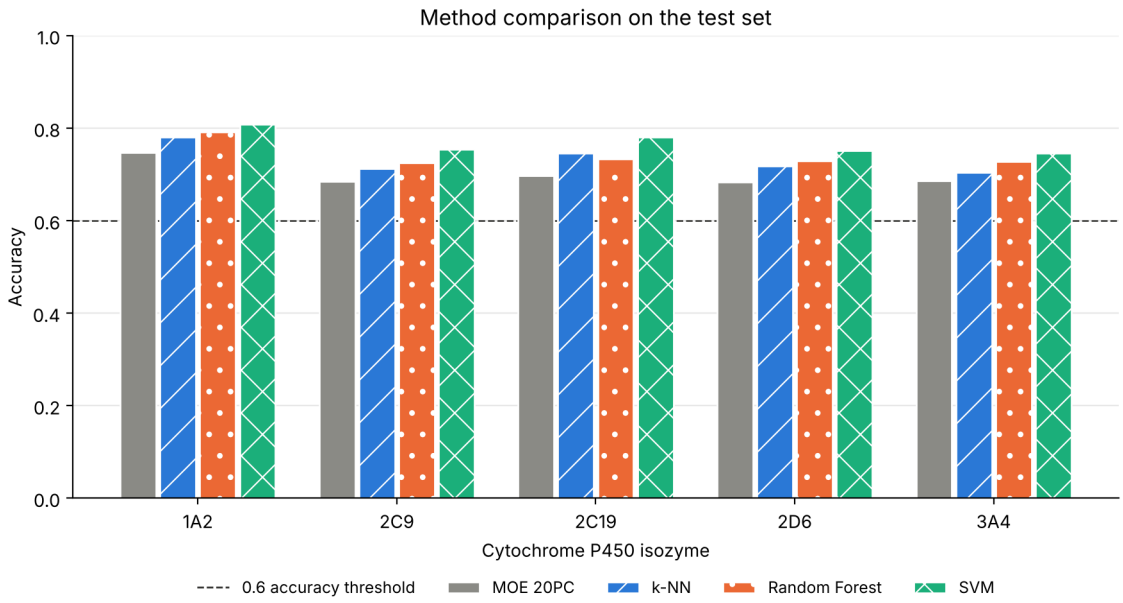


Figure 3.8: Inhibitor Classification Method Comparison on the Test Set

On the test set, all three Python classifiers exceeded MOE Binary QSAR for every isozyme, and the support vector machine classifier had the highest accuracy in each case.

CHAPTER 4

Discussion

4.1 Other Attempts at Modeling PubChem AID 1851

PubChem AID 1851 has been the basis for several attempts to advance *in silico* screening development. Several research groups have developed large-scale single target models for individual isozymes [Vasanthanathan et al., 2009, Sridhar et al., 2012, Sun, 2012, Novotarskyi et al., 2010]. Another group pursued all five isozymes but used a proteochemometric method that also takes into account each isozyme's protein sequence [Lapins et al., 2013]. Finally, Cheng et al. built single-target QSAR models for all five CYP isozymes, the work most comparable to this thesis [Cheng et al., 2011]. All of these models exceeded the 0.6 accuracy threshold used in this thesis.

Cheng et al. modeled the same five isozymes but used a combined classifier on a larger data set of more than 24,700 unique compounds. These compounds also came from PubChem AID 1851 and included the 17,143 used in this study, plus additional compounds from PubChem AID 410. The group used an ensemble of different independent machine learning classifiers including support vector machine, C 4.5 decision tree, k-nearest neighbors, and naive Bayes. The final ensemble was brought together

using a back-propagation artificial neural network (BP-ANN). Those models were also internally validated by 5-fold cross-validation and then evaluated with a test set composed of about 9000 diverse unique compounds previously unseen by the classifier. Test set validation measures ranged from 0.764 to 0.886, overlapping the upper end of the test accuracies reported here (0.683–0.808) and extending above it, with a larger data set and an ensemble method. Cheng et al. claim these classification models are applicable for virtual screening for inhibitors of the five major CYP isozymes or can be used as simple filters of potential chemicals in drug discovery [Cheng et al., 2011].

4.2 Sources of Error

This assay required recombinant sources of CYP enzymes because the probes are not sufficiently P450 isozyme-selective to be used with those derived from human liver microsomes. Another potential source for false positives in these assays can be compounds that interfere with light generation directly, such as compounds that interfere with luciferase enzymatic activity [Zlokarnik et al., 2005].

The results of the large-scale screening of PubChem AID 1851 against five CYP isozymes found that most compounds in a typical chemical library inhibited several isozymes, and only a small fraction inhibited none [Veith et al., 2009].

The uniformly lower accuracy of the MOE models on non-inhibitors than on inhibitors may reflect the composition of the non-inhibitor class, which contains both true non-inhibitors (score 0) and inconclusive compounds (scores 1–39). A binary labelling may discriminate less well than one with more classes. The Python classifiers show no such consistent pattern (Section 3.11). Future models could test the effect of class composition by adding a separate inconclusive class or changing the Activity Score threshold.

4.3 Machine Learning for Pharmaceutical Sciences

> Developing successful machine learning applications still requires a substantial amount of "black art" that is hard to find in textbooks [Domingos, 2012].

Unlike the data in this study, raw data are often not amenable to learning, but features constructed from them can be. Features are the most important factor in a good model. If there are many independent features that correlate well with the class, learning is easy. However, if the class is a very complex function of the features, training a good model may take too long or be impossible [Domingos, 2012]. Most of the effort in a machine learning project goes into feature engineering, an area where domain expertise and experience with chemical library design can mean the difference between success and failure.

Machine learning is not a one-shot process of building a data set and running a learner, but rather an iterative process of running the learner, analyzing results, modifying the data and/or learner, and repeating. Learning is often the quickest part of this, but only because of all the effort that has gone into sharing and codifying prior knowledge. Feature engineering is considered more difficult because it is domain-specific, while learners can by and large be general-purpose [Domingos, 2012].

Domain expertise in feature engineering has no substitute. The alternative, running a learner on a very large number of features to find useful combinations, may be too time-consuming or cause overfitting [Eklund et al., 2014].

As a rule, it pays to try the simplest learners first (e.g. naive Bayes before logistic regression, k-nearest neighbors before support vector machines). More sophisticated learners may be seductive, but they are usually harder to use, because there are more tuning parameters and their internals are more opaque [Domingos, 2012].

One way to divide learning algorithms is as follows: those whose representation has a fixed size, like linear classifiers, and those whose representation can grow with

the data, like decision trees. Fixed-size learners can only take advantage of so much data. Variable-sized learners can in principle learn any function given sufficient data, but in practice may not, because of limitations of the algorithm or computational cost. Also, because of the curse of dimensionality, no existing amount of data may be enough [Domingos, 2012]. Each type of learner comes with its own assumptions and, consistent with the no free lunch theorem, none is best in all situations, although some are clearly better than most [Hand, 2006, Fernández-Delgado et al., 2014].

Perhaps because different learners see the data from different angles, combining them into an ensemble often improves accuracy, and ensembles are now standard in machine learning. Techniques such as bagging, boosting and stacking resample the training data or reweight previously misclassified instances to reduce variance while minimizing bias. The simplest ensemble is a majority vote: each instance is assigned the class predicted by most of the models.

Because ensembles frequently outperform their constituent models, a logical next step is to combine the models developed here.

CHAPTER 5

Conclusion

QSAR models and their value in informing regulatory decision making will likely increase with the standardization of analytical approaches, more complete and reliable data collection methods and a better understanding of toxicity mechanisms in the role of disease as well as individual susceptibility to adverse clinical events [Kruhlak et al., 2012].

I built and compared four methods for classifying compounds as inhibitors or non-inhibitors of five cytochrome P450 isozymes. All methods reached test accuracies between 0.683 and 0.808, well above the 0.5 expected from random guessing on balanced classes. On every isozyme, all three Python methods were more accurate than MOE Binary QSAR, and support vector machines were the most accurate. I did not test whether these differences are statistically significant.

Many other methods remain to be tried, particularly neural networks and other Bayesian methods. The data and code for this study are publicly available, so others can benchmark new methods against these results.

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